

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Principles of pollution control (CPA, CGM, new technologies) - Set 1

Mark scheme

AQA spec 3.4.3.1, 3.4.3.3; Pollution; textbook pp. 243-245, 303-304

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Predicting the route / movement a pollutant will take through the environment (based on its properties and features of the environment) (1) • To assess where it will become concentrated / which locations and organisms are most at risk, so monitoring can focus on those sites (1)	2
01.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Sea current velocity and direction (1) • Properties of the radionuclides: half-life, solubility, tendency to adsorb onto sediments / bioaccumulate (1) • Food chain pathways, eg uptake by seaweed, shellfish, fish eaten by people (1) • Geology / hydrology of the coast; sediment deposition areas (mud flats) (1) • Human activities in the area: fishing, seaweed collection, beach use (1)	3
01.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Identify the group of people whose lifestyle gives them the highest exposure (eg people who eat a lot of locally caught shellfish / fish, live near the mud flats, collect seaweed) (1) • Monitor their exposure / the concentration of radioactive material in the foods they eat or their body (1) • If their exposure / risk is acceptably low, everyone else is assumed to be at even lower risk; if not, emissions are controlled (1) • CGM assesses exposure before health effects occur (not used to detect damage) (1)	3

Question	Marking guidance	Marks
01.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Seawater (1) • Sediment / mud (1) • Seaweed (1) • Shellfish / molluscs / fish (1) • Milk, vegetables or meat from nearby farms (1) • Atmospheric dust (1) • Soil (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243, 303-304; AQA spec 3.4.3.1.	
02.1	<i>1 mark for each point.</i> • Location: discharge into the sea / large water bodies where currents dilute and disperse the pollutant (1) • Location: chimneys / emissions sited downwind of urban areas so pollutants are not carried over populations; avoiding valleys and enclosed water bodies where pollutants concentrate (1) • Location: not discharging wastes onto permeable rock above an aquifer / not siting landfill on aquifers (1) • Timing: restrict polluting activities (industrial combustion, vehicle use) during temperature inversions when dispersal is poor (1) • Timing: discharge liquid effluent when river flow is high / on an ebb tide so it is diluted and carried away (1) • Timing: pesticide spraying at night / on still days; noisy operations restricted to daytime; night-flight restrictions (1)	6
02.2	<i>1 mark for each point.</i> • The organisation / person that causes the pollution is responsible for (paying for) the damage caused (1) • If the cost of preventing the pollution is less than the cost of the damage / fines, there is a financial incentive to prevent it; works best when penalties are high and escape is unlikely (1)	2
02.3	<i>1 mark for each point.</i> • It is assumed that the waste will cause pollution / harm until research shows it is unlikely to (1) • So release is not permitted until it has been tested; lack of knowledge of a problem is not an excuse / does not reduce responsibility (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
03.1	<i>1 mark for each point.</i> • Cost increases as the percentage prevented increases (1) • The relationship is not linear: cost rises slowly at first then very steeply / exponentially above about 80-90% (1)	2
03.2	<i>1 mark for the correct answer.</i> • $16.0 - 5.0 = 11.0$ (£ million) (1)	1

Question	Marking guidance	Marks
03.3	<i>1 mark for each point.</i> • ALARA: emissions should be As Low As Reasonably Achievable (1) • It is rarely practical / affordable to reduce emissions to zero; low-level emissions may have acceptable impacts so an acceptable emission level is set (1) • BATNEEC: Best Available Technology Not Entailing Excessive Cost - choosing the most effective control equipment whose cost is not excessive (1) • The factory would install equipment that gives a high level of control (eg 80-90%) rather than paying £16 million for 99% (1)	4
03.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Industry / city may have expanded so there are more pollutant sources and total emissions have risen (1) • Better understanding of the health / environmental effects so stricter standards are set (1) • Improved technology becomes available (BATNEEC changes) / cost of better control falls (1) • Pollutant may have accumulated (persistent) so background levels are higher (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 244-245; AQA spec 3.4.3.1.	
04.1	<i>1 mark for each point.</i> • Production prevention: desulfurisation of fuels before combustion; low-temperature (fluidised bed) combustion to prevent NO_x; smokeless fuels (1) • Prevention of release: electrostatic precipitators / bag filters / cyclones for smoke; catalytic converters; flue gas desulfurisation; bund walls; landfill liners (1) • Post-release remediation: oil spill clean-up (booms, skimmers, dispersants, bioremediation); phytoremediation of metal-contaminated land; liming of acidified lakes (1) • Alternative processes: electric vehicles instead of petrol / diesel; renewable energy instead of fossil fuels; pyrethroids instead of organochlorines; fuel cells instead of internal combustion engines (1)	4
04.2	<i>1 mark for each point.</i> • Remediation is often incomplete / cannot recover all the pollutant (eg oil spills, dispersed pollutants), so some damage still occurs (1) • Prevention avoids the damage entirely and is usually cheaper than clean-up / avoids polluter-pays costs; moving from post-production treatment to non-release is a circular economy principle (1)	2
04.3	<i>1 mark for each point.</i> • Satellites detect and track the extent / movement of oil slicks over large areas, quickly and repeatedly, so clean-up can be directed and polluters identified (1) • GPS tracking monitors ship positions so navigation errors / collisions / groundings are reduced and offshore shipping routes are enforced (1) • Polymers with a high surface area adsorb metal ions from contaminated water / effluent so they can be removed and recovered (1) • Plants that absorb heavy metals (hyperaccumulators) are grown on contaminated land, harvested and disposed of / metals recovered, cleaning the soil (1)	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 244, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	
05.1	<i>Correct answer only.</i> • C (1)	1
05.2	<i>1 mark for each point.</i> • Monitoring can be focused on the sites predicted to be most at risk / where the pollutant will concentrate (1) • No need to sample sites that were never likely to be polluted; sampling also checks the accuracy of the predictions (1)	2
05.3	<i>1 mark for each point.</i> • Behaviour of new chemicals can be predicted from similar existing chemicals: it is likely to persist, bioaccumulate and biomagnify (1) • So acceptable emission limits, control methods and monitoring programmes can be set before release (precautionary principle) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
06.1	<i>Correct answer only.</i> • A waste is assumed to cause pollution until research shows it is unlikely to do so (1)	1
06.2	<i>1 mark for each point.</i> • If the cost of preventing damage is less than the (penalty for) damage there is a financial incentive to prevent pollution (1) • If polluters can escape their responsibilities / penalties are low, there is little incentive to comply (1)	2
06.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Marine discharges where currents dilute and disperse emissions (1) • Emissions downwind of urban areas (1) • Not discharging waste onto permeable rock above an aquifer (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-244; AQA spec 3.4.3.1.	
07.1	<i>1 mark for each point.</i> • Natural processes may concentrate the material (eg adsorption onto sediments, bioaccumulation and food-chain concentration) (1) • It may be carried to locations where it is hazardous, such as fishing grounds or beaches used by people (1)	2
07.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Physical state of the effluent (solution / solid) (1) • Ocean currents / river flow (1) • Effect of pH and oxygen on solubility (1) • Bioaccumulation and food-chain concentration (1) • Food sources and consumption by local people (eg shellfish, seaweed) (1) • Half-lives of the isotopes (1) • Meteorological conditions for any gaseous release (1) • Nature of the environment - geology, vegetation (1)	4
07.3	<i>1 mark for each point.</i> • Monitoring focuses on the sites most at risk of contamination (1) • No need to sample sites that were never likely to be polluted; sampling checks the accuracy of the predictions (1)	2

Question	Marking guidance	Marks
07.4	<i>1 mark for each point.</i> • It assesses the risk to the people most exposed because of their lifestyle before any health effects occur (1) • If their risk is acceptably low everyone else has a lower risk; if unacceptable, emissions are reduced to prevent future exposure (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	
08.1	<i>1 mark for the correct answer.</i> • $16.0 - 5.0 = \text{£}11 \text{ million}$ (1)	1
08.2	<i>1 mark for each point.</i> • It is rarely practical to reduce emissions to zero and the impact of low-level emissions may be acceptable, so an acceptable level must be decided (1) • ALARA: emissions should be As Low As Reasonably Achievable (1) • BATNEEC: the Best Available Technology Not Entailing Excessive Cost is chosen to achieve this (1)	3
08.3	<i>1 mark for each point.</i> • As industries expand or cities grow the number of pollutant sources increases (1) • So the total emissions / combined concentration rises even if each source is unchanged, and higher standards are needed (1)	2

Question	Marking guidance	Marks
08.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Critical Pathway Analysis: predicting movement of mobile persistent pollutants (state of effluent, wind, currents, geology, pH / oxygen and solubility, bioaccumulation, food sources, half-lives) so monitoring focuses on sites at risk; mainly radioactive discharges; checked by sampling • Critical Group Monitoring: exposure of the most-exposed group monitored before health effects; if their risk is acceptable everyone else is safer; emissions reduced if not • Emission control strategies: location (marine discharges, downwind of cities, not over aquifers) and timing (avoiding temperature inversions, low river flow) • Principles: polluter pays (incentive depends on penalties and enforcement), precautionary principle (assume harm until shown otherwise; ignorance no excuse), ALARA / BATNEEC and the non-linear cost-efficiency relationship • Selection of methods: production prevention (desulfurisation), prevention of release (precipitators, catalytic converters), post-release remediation (oil spill clean-up, phytoremediation), alternative processes (electric vehicles, renewables, pyrethroids) • Evaluation: prevention usually cheaper and more effective than remediation; costs rise steeply near 100%; CPA relies on understanding of pathways; enforcement and international agreement needed for transboundary pollutants; judgement about acceptable risk is political as well as scientific • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	